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PROJECT TITLE: BOUTIQUETRACK; AN ONLINE BOUTIQUE INVENTORY MANAGEMENT SYSTEM FOR RETAIL BUSINESSES

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CONCEPT NOTE.

1.0 Introduction

Juba, the capital city of South Sudan, has experienced rapid growth in the retail sector, particularly in the fashion and clothing industry. This growth has been driven by urbanization, increased disposable income, and the entrepreneurial spirit of young people. Major markets such as Konyo Konyo, Custom Market, and Juba Market now host hundreds of boutiques selling clothes, shoes, and accessories. Despite this expansion, the operational practices of most retail shops remain largely manual. Product records, sales transactions, and stock levels are typically recorded in exercise books, receipts are handwritten, and stock-taking is done physically at the end of each month. This manual approach is not only time-consuming but also prone to human errors, data loss, and theft. As a result, shop owners often lack visibility into their actual stock levels, sales trends, and profitability, which hinders effective decision making.

Globally, the adoption of Information and Communication Technology has transformed retail operations by improving accuracy, speed, and accountability. In South Sudan, however, the adoption of such technologies among SMEs remains low due to cost, complexity, and lack of localized solutions. With the increasing availability of computers and affordable internet access in Juba, there is a clear opportunity to introduce simple digital solutions tailored to the local context. This concept note therefore proposes Boutique Tracking system, a web-based boutique management system designed specifically for retail shops in Juba. The system will allow different shops to register and manage their data independently in a secure, multi-tenant environment. By automating inventory management, sales processing, and report generation, BoutiqueTrack aims to promote efficiency, transparency, and professionalism in the retail sector.

1.1 Background

The contribution of Small and Medium Enterprises to the economy of South Sudan cannot be overstated. According to the Ministry of Trade and Industry, SMEs account for over 70% of businesses in the country and are a major source of employment for youth and women. The retail sub-sector in particular has grown rapidly in Juba due to increased urban migration and improved consumer purchasing power. The city's central markets are now filled with boutiques selling a wide range of products to meet the demands of a growing middle class.

However, despite their economic importance, SMEs in South Sudan face numerous challenges that limit their growth and competitiveness. One of the most critical challenges is poor business management and record-keeping practices. A study by UNDP on SMEs in South Sudan revealed that over 80% of small retail businesses still rely entirely on manual systems for managing their operations. The report further noted that lack of access to affordable technology and lack of technical skills are the main barriers to ICT adoption among SMEs in the country.

Literature from other developing countries shows similar patterns. Olatokun and Igbinedion in their 2019 study on SMEs in Africa observed that manual record-keeping leads to frequent data loss, inaccurate financial reporting, and poor inventory control. They concluded that businesses using manual systems are 40% more likely to experience stock discrepancies and revenue leakage compared to those using automated systems. In the context of Juba, this problem is compounded by high staff turnover. When records are kept in books, it becomes difficult for a new staff member to trace past transactions or understand stock movement.

Another issue documented in literature is the lack of business intelligence. Laudon and Laudon in Management Information Systems argue that one of the primary benefits of information systems is the ability to transform raw data into useful information for decision making. In most Juba boutiques, however, such transformation does not happen. At the end of the month, owners struggle to produce a simple sales report or profit and loss statement. This makes it difficult to plan for restocking, set prices, or identify profitable product lines.

Globally, the retail industry has been transformed by ICT. According to Turban et al., modern retail systems use databases to track thousands of products with different attributes, process sales in seconds, and generate management reports automatically. Systems like ERP and cloud POS have helped large retailers improve efficiency and reduce operational costs. However, these systems are expensive, complex, and require stable internet and IT support. They are therefore not suitable for small boutiques in Juba that operate with limited capital and basic computer skills.

The situation in South Sudan is further constrained by infrastructure challenges. Although internet connectivity has improved in Juba, it is still intermittent and expensive. Power outages are also common. As a result, any digital solution proposed for this context must be simple, offline-capable on a local server, and require minimal maintenance.

Given this background, there is a clear need for a context-specific, low-cost, and easy-to-use digital solution. The proposed Boutique Tracking system seeks to fill this gap by providing a web-based application that can run on a local XAMPP server, does not require constant

internet, and is designed around the actual workflows of Juba boutiques. By introducing such a system, this study contributes to the broader national agenda of promoting ICT adoption and digital transformation of SMEs as outlined in the National SME Policy Framework of South Sudan.

1.2 Problem Statement

Retail shop owners in Juba face several operational challenges that stem from the continued reliance on manual systems. The first major challenge is inefficient inventory management. Retail products often come with multiple attributes such as size, color, and brand. Tracking these variations using paper records is difficult and frequently leads to stockouts of fast-moving items or overstocking of slow-moving ones. This directly affects sales and cash flow. Secondly, the sales process is manual and error-prone. Staff must calculate totals manually, write receipts by hand, and then update stock levels later. This process consumes time during peak hours and increases the likelihood of calculation mistakes and discrepancies between physical stock and recorded stock.

Thirdly, there is a critical lack of business insights. Because records are kept in books, it is difficult for owners to generate timely reports on daily sales, monthly revenue, or top-selling products. Without this data, owners make decisions based on guesswork rather than evidence. Fourthly, existing commercial solutions such as ERP and POS systems are designed for large enterprises. They are expensive to purchase, require technical expertise to maintain, and are not customized for the needs of small retail shops in Juba. As a result of these challenges, there is a pressing need for a simple, affordable, and locally relevant web-based system that can be adopted by different retail shops to streamline operations and support growth.

1.3 Main Objective

The aim of this project is to design and develop BoutiqueTrack, an online management system for retail shops in Juba. The system will automate key business processes including product management, sales processing, and inventory control. By doing so, it seeks to improve operational efficiency, reduce errors, and provide shop owners with timely business information to support better decision making.

1.3.1 Specific Objectives

i. To study and document the current business processes of retail shops in Juba through interviews, questionnaires, and direct observation to understand user needs, challenges and pain points.

ii. To design and develop a web-based retail management system with modules for product management, sales/POS, inventory, user management with role-based access for admin and cashier, and a relational database to support product variations and data integrity.

iii. To implement automated report generation for daily, weekly, and monthly sales and inventory data to support decision-making and replace manual record-keeping.

iv. To test and evaluate the system with a selected retail shop in juba through user acceptance testing to validate its functionality, usability, and effectiveness in improving business operations.

1.4 Scope of the Project

i. Geographical Scope

This study will be conducted in Juba, the capital city of South Sudan. Specifically, the research will focus on one retail boutique located within Juba's central business markets such as Konyo Konyo or Custom Market. The selected shop will serve as the case study for requirement analysis, system design, implementation, and evaluation. While the system is being developed for one shop, the findings and design may be generalized to other similar retail shops operating within Juba due to the similarity in business processes and challenges.

ii. Functional requirements scope

The BoutiqueTrack system will be designed to automate the core business processes of a single retail shop. First, it will include User Management with secure login and authentication for two roles, Admin and Cashier, to ensure that only authorized personnel can access the system. Second, it will provide Product Management which will allow users to add, edit, delete, and view products. Each product will have details such as name, category, size, color, brand, buying price, selling price, and quantity. Third, the system will have a Sales and POS Module to process customer transactions. This module will calculate subtotals and discounts, print receipts, and automatically deduct sold items from the inventory. Fourth, it will include Inventory Management for real-time tracking of stock levels. The system will also allow stock adjustments and will generate automated low-stock alerts to improve stock tracking efficiency and prevent stockouts. Fifth, the system will have a Reporting Module that will generate automated daily, weekly, and monthly reports. These reports will cover sales data, inventory levels, and product performance to support decision-making. Sixth, the system will include a Dashboard

that will display key business metrics. These will include total sales made today, total revenue earned this month, top-selling products, and items that are running low on stock. To maintain focus, the following features are excluded from this version: multi-branch support, e-commerce for online customers, barcode or QR code scanning, integration with mobile money or online payment systems, and the development of a mobile application. These features are recommended for future expansion.

iii. Non-Functional Requirements Scope

The system will also be developed to meet specific quality attributes to ensure usability and reliability. The system will prioritize Usability by having a simple, intuitive, and user-friendly interface that can be used by staff with basic computer skills. For Performance, the system will respond to user actions and generate reports in less than 5 seconds under normal load.

Reliability will be ensured through data accuracy and consistency using proper database constraints and transaction management. Security will be maintained by restricting access to the system through username and password authentication with role-based permissions. The system will be Maintainable because it will be built using standard and well-documented technologies to allow easy updates and future enhancements. Finally, Portability will be achieved since the system will be web-based and accessible on any device with a modern web browser and a local XAMPP server.

1.5 Significance of the Study

i. To Retail Shop Owners and Staff

The BoutiqueTrack system will provide shop owners with a cost-effective tool to automate daily operations. By automating product, sales, and inventory management, the system will reduce human errors, minimize stock losses, and save time spent on manual recording and calculations. Staff will benefit from faster billing through the POS module and clearer work responsibilities through role-based access. This will lead to improved productivity and better customer service delivery.

ii. To Customers

Customers will benefit from quicker service at the checkout point and accurate billing. The ability to print receipts will increase transparency and build trust

between the shop and its customers. Faster service during peak hours will also improve the overall shopping experience.

iii. To the Government and NGOs Supporting SMEs

The Government of South Sudan and development partners such as UNDP are promoting the digitalization of SMEs to support economic growth and job creation. This study demonstrates a practical, low-cost model for digitizing retail businesses in Juba. If adopted, similar systems can be replicated across other sectors to improve business efficiency and contribute to formalization of the informal economy.

iv. To the Academic Community and Future Researchers

There is limited literature and few practical ICT solutions developed specifically for SMEs in South Sudan. This study will contribute to academic knowledge by providing a documented process of analyzing, designing, and implementing a context-specific management system. It will serve as a reference material for students and researchers interested in ICT4D and business systems in developing countries.

v. To the Developer/Researcher

For the researcher, this project provides hands-on experience in the full software development life cycle, including requirement gathering, system design, implementation using PHP and MySQL, testing, and deployment. It strengthens practical skills in full-stack web development and systems analysis.

1.6 Research Questions

i. Who are the key users of a boutique management system in Juba and what are their roles in daily shop operations?

ii. What are the current challenges faced by retail shops in Juba in managing inventory, sales, and business reporting?

iii. What are the functional and non-functional requirements needed for an effective web-based boutique management system for a retail shop in Juba?

iv. How can a web-based system improve stock tracking efficiency and reduce errors in sales and inventory management?

v. How can automated reports and a dashboard support better decision making for a retail shop owner?

vi. To what extent is the developed BoutiqueTrack system usable and effective in improving the operations of a retail shop in Juba?

1.7 JUSTIFICATION OF THE STUDY

The justification for this study is based on three key grounds: operational need, economic impact, and academic contribution.

First, operational need, the continued use of manual systems in retail shops in Juba has resulted in persistent challenges such as stock losses, inaccurate sales records, and poor customer service. Literature shows that manual record-keeping increases the risk of errors by up to 40% and leads to poor inventory control in SMEs. Given the competitive nature of Juba's retail market, there is an urgent need to automate these processes to reduce errors and improve efficiency.

Second, economic impact, SMEs are the backbone of South Sudan's economy and contribute to employment and household income. However, their growth is hindered by lack of access to affordable technology. Developing a low-cost, web-based system like BoutiqueTrack will enable shop owners to manage their businesses professionally, reduce operational costs, and increase profitability. This aligns with the governments and UNDP's goal of promoting digitalization of SMEs to support economic recovery and resilience in South Sudan.

Third, academic contribution, most existing management systems are designed for large corporations and are not affordable or suitable for the local context. There is limited research and few working prototypes of ICT solutions tailored for SMEs in South Sudan. This study will therefore contribute to academic literature by providing a practical model for developing context-specific business systems. It will also serve as a reference for future researchers and developers working on digital solutions for SMEs in developing economies.

In summary, this study is justified because it addresses a real problem, supports economic development, and fills a gap in local academic research.

1.8 Assumptions and Limitations

The study assumes that target retail shops have access to at least one computer and intermittent internet connectivity for system access. It also assumes that shop owners and staff are willing to be trained on using the new system. The main limitations of the project include the lack of an offline mode, meaning the system requires a local server to be running. The system will also rely on manual data entry and will not support barcode scanning in this version. Security will be limited to role-based access and will not include advanced features such as two-factor authentication.

1.9 **Methodology**

The project will adopt the Waterfall Model for system development because the requirements are clear, the scope is well defined, and sequential execution is suitable for an academic project. The first phase will involve requirement gathering through semi-structured interviews with three retail shop owners in Juba and direct observation of their daily operations. Data collected will be analyzed to identify functional and non-functional requirements. The second phase will focus on system analysis and design. During this phase, Entity Relationship Diagrams, Data Flow Diagrams, and user interface mockups will be developed to guide implementation.

The third phase will involve actual system development. The backend will be developed using PHP 8, the database will be implemented in MySQL, and the frontend will be built using HTML5, CSS3, Bootstrap 5, and JavaScript. The fourth phase will involve system testing. Unit testing will be conducted to verify individual modules, followed by integration testing and user acceptance testing with selected shop owners to gather feedback. The final phase will involve system deployment on a local server using XAMPP and training of end users to ensure effective adoption. The primary tools for development will include XAMPP, Visual Studio Code, MySQL Workbench, and http://Draw.io.

1.10 Expected Outcomes

At the end of this project, a fully functional web-based system named BoutiqueTrack will be delivered and deployed. The system will be capable of supporting multiple retail shops with separate accounts, roles, and data. It is expected that the system will reduce the time spent on manual recording and report generation by at least 50 percent. The accuracy of stock and sales records is expected to improve significantly due to automation and real-time updates. Shop owners will have access to timely and accurate business information, which will enable them to make informed decisions regarding purchasing, pricing, and promotions. Furthermore, the project will contribute to the broader goal of digitalizing SMEs in South Sudan and improving their competitiveness.

1.11 Conclusion

In conclusion, BoutiqueTrack addresses a real and pressing need in the retail sector of Juba. The continued reliance on manual processes is no longer sustainable given the growth and competitiveness of the market. By providing a simple, affordable, and secure digital solution that different shops can use independently, BoutiqueTrack will enable small businesses to operate more efficiently, minimize losses, and make data-driven decisions. Ultimately, the system will contribute to the growth and professionalization of retail SMEs in South Sudan.